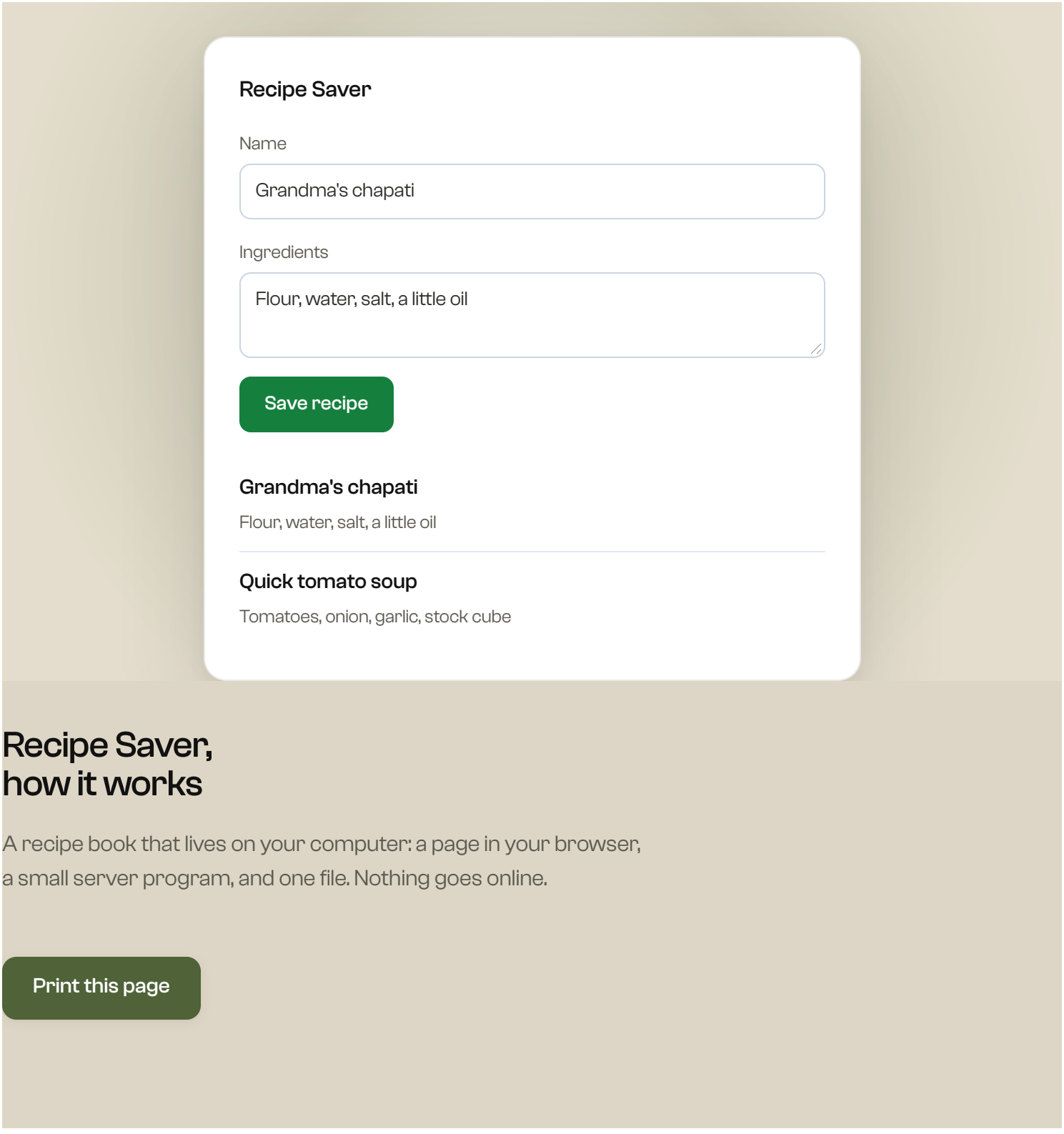


Recipe Saver, how it works

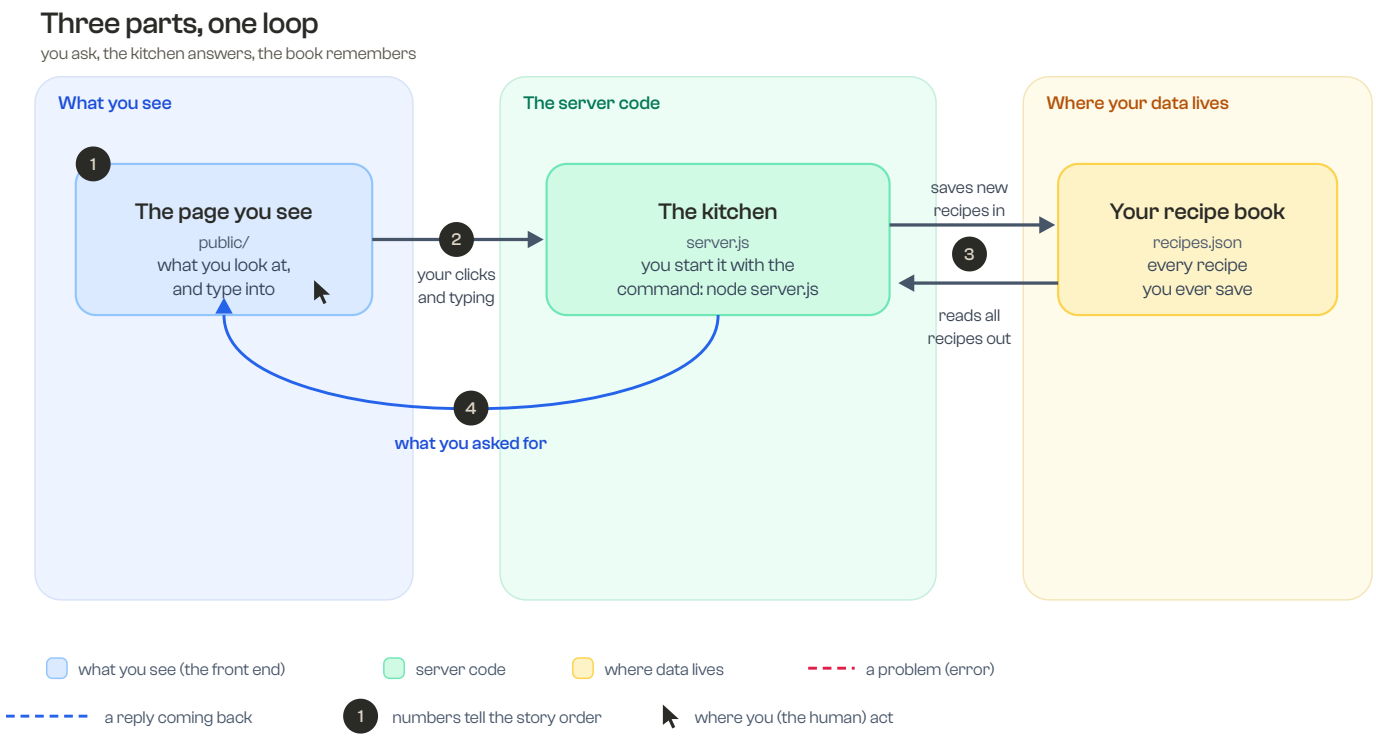
A recipe book that lives on your computer: a page in your browser, a small server program, and one file. Nothing goes online.

Print this page

The big picture

Three parts, one loop. The page in your browser is the table: a form, a list, a search box. Behind it, a small program called the **server** (the program that listens for requests and answers them) is

the kitchen: it checks every recipe you save and keeps the book. The book itself is one file, **recipes.json**. Once you can read this picture, every other one below is a zoom-in.

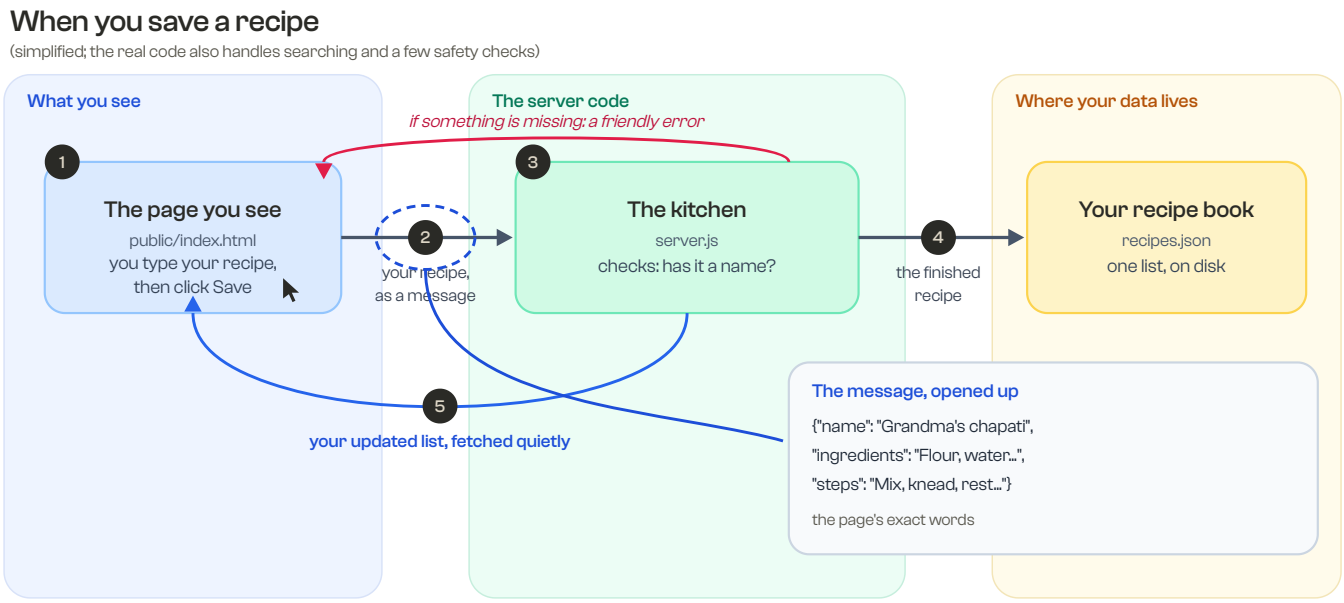


In plain words: 1 The page is your table, where you look, click and type. 2 Everything you do is carried to the kitchen: your server, server.js, which you start with the command node server.js. 3 The kitchen writes new recipes into your book, recipes.json, and reads all recipes back out. 4 The answer returns to your page. It is a restaurant at your own desk: menu in front of you, kitchen behind the wall, recipe book in the kitchen. Every click is one loop of ordering and serving.

What happens when you save

You fill in the name, maybe the ingredients and steps, and click Save. Five things happen in about a blink, and every one of them passes through the kitchen.

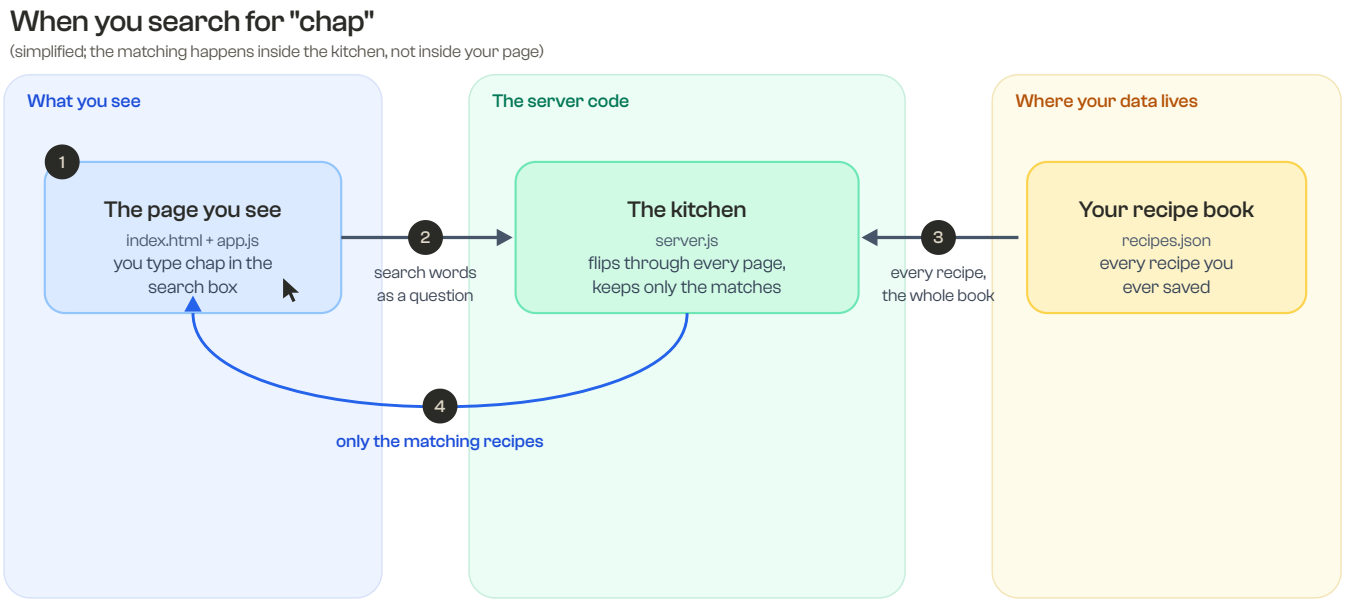




In plain words: 1 You type your recipe and click Save. 2 The page wraps what you typed into a message; the inset shows exactly what it looks like on its way over. 3 The kitchen does one quality check: does it have a name? If something is missing, the red arrow sends a friendly error straight back to your form and nothing gets saved. 4 If the recipe passes, the kitchen writes it on a fresh page of your book, next to everything you saved before. 5 Then, quietly and without you doing anything, your page asks for the whole book again and redraws the list. That is why your new recipe appears instantly.

What happens when you search

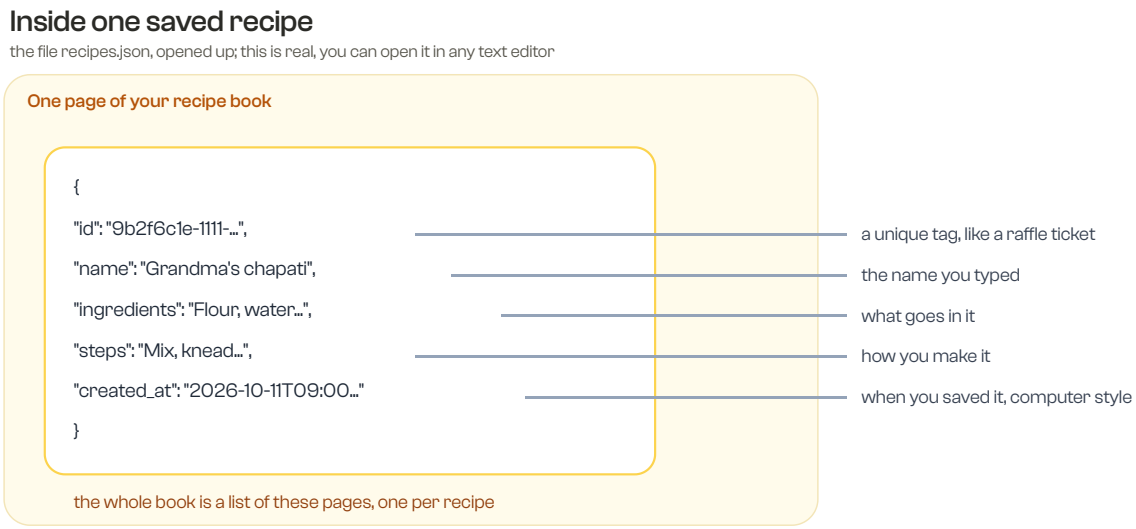
Every letter you type in the search box triggers the same little trip. Here it is with the word chap.



In plain words: 1 You type chap in the search box. 2 The page sends your search words to the kitchen as a question: any recipes with chap in the name or the ingredients? 3 The kitchen pulls out the whole book and flips through every page, keeping only the matches. 4 Only the matching recipes come back, and your list redraws with just those. The search is not magic: the kitchen reads your entire book every single time and simply skips what does not match. That is why it feels instant with ten recipes, and why a million-recipe book would need a smarter kitchen.

Where your data lives

Your book is a plain text file, recipes.json, sitting next to the code. You can open it in any text editor. Here is one page of it, opened up and translated.



In plain words: one saved recipe is a little form with five boxes. You filled in three: the name, the ingredients, the steps, exactly as you typed them. The kitchen stamped the other two for free: a unique tag, so no two pages in the book can ever get mixed up, and the exact moment you saved it. The curly braces and quotation marks are just the book's grammar, a way of writing forms in plain text that both the page and the server can read. The internet's dialect for this is called **JSON**, and yes, people pronounce it Jason.

The book over time

Four states, one file: what your book looks like from the day you start it to the day a recipe loses its name. These are the app's list, as you would really see it.



Empty

No recipes yet. Your book is empty.

One recipe

Grandma's chapati

saved just now

Many recipes

Grandma's chapati

Quick tomato soup

Coconut curry

Error

A recipe with no name

was not saved

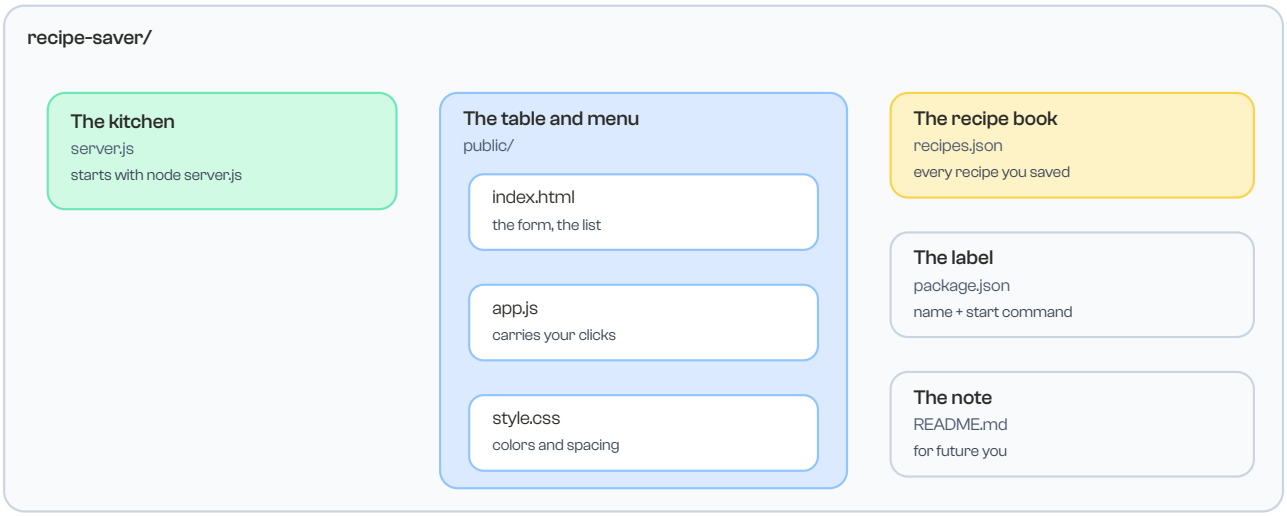
Please give your recipe a name.

In plain words: read left to right: an empty book, your first recipe, a growing collection, and what happens when a recipe arrives without a name. The kitchen refuses it politely and nothing is

saved. The book only ever grows through step 4 of the save story: one file, one list, every state you will ever meet.

The cast of characters

Every file in your project, nested the way it really lives on disk. One human sentence each, including what happens if it goes missing.



In plain words: one folder holds everything. **server.js** is the kitchen: delete it and nothing answers. **index.html** is the page itself; **app.js** the waiter: delete it and the page loads but clicks do nothing; **style.css** the decor: delete it and everything still works, just plain. **recipes.json** is the book, the only place your recipes exist, so copy it somewhere safe if you love them. **package.json** is the label on the box; **README.md** the note for future you.

Words you'll hear

Every technical word this document used, in the order you met it.

browser

The program you are reading this in: Chrome, Safari, Firefox. It opens pages and talks to servers for you.

server

The program that listens for requests and answers them. Yours is server.js, and it runs on your own computer, not on the internet.

page / front end

Everything you can see and click: the form, the list, the search box. The files in public/. Front end is simply the tech name for the part you see.

request / response

One trip: a question your page sends to the server, and the answer that comes back. Every click silently sends one.

API

The kitchen's service window: the fixed menu of messages the server understands. Your kitchen understands three: show me the recipes, save this recipe, search for this word.

JSON

Jason. The plain-text grammar for writing little labeled forms, the one you saw inside a saved recipe. The whole book, and every message between page and kitchen, is written in it.

localhost / port

Your own computer's nickname in the browser. localhost:3000 means this same machine, door number 3000. The port is just a numbered door, and the kitchen listens at door 3000.

validation

The kitchen's quality check before cooking: does this recipe have a name? The polite red error you saw is validation talking.

unique id

A made-up tag, long and random, that no two recipes share, like a raffle ticket stapled to each page of the book.

400 and 404

The kitchen's numbered ways of saying no. 404: never heard of that dish (wrong address). 400: this order doesn't make sense (something was missing or unreadable).

How to explain it to a friend

"Recipe Saver is a digital recipe book that lives on my computer. A page in my browser is the table, a small program called the server is the kitchen, and a single file is the book. When I save a recipe, the kitchen checks it, glues it into the book, and reads the whole book back to my page. When I search, the kitchen flips through every page and serves up only the matches. Nothing goes online. It's all mine."

Explained with Explain It With Pictures, a free open-source skill for AI coding agents.

Photographs: "Woman Cooking Kitchen" by U.S. National Archives (CC0, via rawpixel); "Curry powder in the spice-bazaar in Istanbul" by Thomas Steiner (CC BY-SA 3.0, via Wikimedia Commons).